

AI on the Road: A Comprehensive Analysis of Traffic Accidents and Autonomous Accident Detection System in Smart Cities

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Abstract—Accident detection and traffic analysis are critical components of smart city and autonomous transportation systems that can reduce accident frequency and severity and improve overall traffic management. This paper presents a comprehensive analysis of traffic accidents in the United States using data from the National Highway Traffic Safety Administration (NHTSA) Crash Report Sampling System (CRSS). This study analyzed traffic accident trends, including accidents, accident severity in different regions across the US, and the percentage distribution of different types of collisions. To address the challenges of accident detection and traffic analysis, this paper proposes a framework that uses traffic surveillance cameras and action recognition systems to detect and respond to traffic accidents quickly and accurately. The proposed framework harnesses the power of traffic cameras, data processing, and machine learning algorithms to create an efficient solution for identifying and responding to traffic accidents. The system's integration with emergency services and first responders can reduce the response time of law enforcement officers and eliminate reporting errors. The use of advanced technologies, such as the proposed framework, and accident detection systems in smart cities can improve traffic management and reduce the risks associated with accidents. This system can potentially save more human lives and property in the United States. Overall, this study provides valuable insights into traffic accidents in the US and presents a practical solution that can enhance the safety and efficiency of transportation systems.

Index Terms—Traffic Surveillance, Accident Detection, Action Recognition, Smart City, Autonomous Transportation.

I. INTRODUCTION

Road traffic accidents are a significant public health concern, accounting for many fatalities and injuries globally each year [1]. World Health Organization (WHO) report approximated that 1.35 million people die annually from road traffic accidents, and over 50 million suffer non-fatal injuries in road accidents [2]. Road accidents have significant financial implications and economic losses due to loss in productivity, medical expenses, and property and infrastructure damages in the accident vicinity. At the global level, adequate steps have yet to be taken to address the catastrophic effect of road traffic accidents on the global population and public health safety. Some measure put in place in developing countries includes wearing helmets, enforcing speed limits, and penalizing traffic offenders. One of the most preventable causes of death worldwide is road traffic accidents. Despite efforts to reduce the

incidence of road traffic fatalities, low-income countries are impacted mainly by the rise of road accidents due to the lack of infrastructure and emergency services in such events. While high-income countries have seen slow progress in reducing road traffic accidents over the past decade, reports still indicate that 80% of traffic accidents occur in developing countries [3]. The adoption of the United Nations Decade of Action for Road Safety (2021-2030) in improving global road safety with the ambition of preventing road traffic deaths and injuries by 50% presents an opportunity to address this problem [3]. Other research works have demonstrated the ability to reduce road accident fatality significantly by 40% and save over 200,000 lives in post-crash care through evidence-based risk factor intervention [4], [5]. Andres et al. [4] compiled country-specific variables in over 185 countries in providing road traffic safety risk factors. Their research demonstrates that effective road safety measures can reduce the incidence of road traffic fatalities across all countries. In recent years, technological advancements have led to the development of autonomous accident detection systems. These systems are designed to detect accidents rapidly, which can reduce emergency response times, remove reporting errors, and save more lives [6]. Based on the increasing popularity of these systems, there is growing interest in utilizing cost-effective systems in monitoring and detecting traffic accidents [7].

The effectiveness of these systems remains to be determined, despite their potential benefits [8]. In developing and implementing these systems, there are several challenges, including a lack of standardized approaches to accident detection, varying quality of the systems, and the need for data sharing and collaboration. This paper aims to provide an overview of the impact of autonomous accident detection systems, analyze accident hot spots and analyze the secondary effects of traffic accidents on human and economic costs related to traffic accidents. Furthermore, this paper proposed a novel framework for detecting accidents based on action recognition.

II. CURRENT ACCIDENT DETECTION TECHNIQUES

Traffic accidents are a significant cause of death and injury worldwide [3]. In recent years, various techniques have been developed to improve traffic safety and reduce the number

of accidents. One such technique is using sensors for traffic monitoring and accident detection. These sensors can detect changes in traffic flow, detect accidents, and even provide data for predicting future traffic conditions [6], [9]. Khalil et al. [9] proposed using ultrasonic sensors for automatic road accident detection. The proposed system consists of two ultrasonic sensors for measuring distance and sound waves that are used for detecting collisions with obstacles. Most techniques adopted for prompt monitoring of road accidents are expensive and sophisticated. The use of autonomous vehicles and intelligent transportation systems in smart cities can reduce accidents and enhance road safety. The National Safety Council reported a 19.% increase in car fatality in some states across the US in the first half of 2022 as compared to 2021 [10]. The World Health Organization (WHO) reports that 1.35 million people die every year in road accidents, and about 50 million people get injured [9]. Modern-day technology such as surveillance cameras, Global Positioning Systems (GPS), Edge AI, and IoT can be beneficial in developing and deploying deep learning algorithms for detecting and predicting accidents. Accident detection is an important area of research in transportation engineering and road safety. With the rapid advancement in technology, various methods have been proposed for accident detection, ranging from traditional human reporting to modern automated systems that utilize sensors and computer vision. This section will discuss some of the current literature and methodologies used for accident detection.

A. Human Reporting

There is often a misconception that human error is responsible for accidents. However, the situation is often beyond the driver's ability to react [11]. Human reporting has been an essential source of accident data for decades. People involved in accidents or witnessing accidents can provide valuable information that can be used to develop accident prevention strategies. In this method, accidents are reported by eyewitnesses or other individuals who witness the accident. However, depending solely on human reporting can be unreliable and time-consuming. As a result, it may not be suitable for time-critical accident detection. Therefore, using computer vision techniques to detect accidents in real-time automatically is an area of active research. These computer vision systems use cameras and machine learning algorithms to analyze video data and detect accidents as they occur. The combination of sensors, IoT devices, human reporting, and computer vision has the potential to significantly improve traffic safety and accident prevention [6], [12].

B. Sensors

Sensors such as accelerometers, GPS, and gyroscopes can be installed in vehicles to monitor driving behavior and detect accidents. These sensors can detect sudden changes in speed, direction, or orientation, which may indicate an accident. The data collected from these sensors can be processed by algorithms to detect and classify accidents [9]. Several studies have explored using different types of sensors for accident detection. Syedul et al. [13] proposed the use of a GPS

receiver to monitor the speed of a vehicle and detect accidents based on the monitored speed. White et al. [14] research combines sensors and context data for accident detection, relying on smartphone sensors and web services to provide situational reports in traffic accidents to reduce the time taken to alert first responders about traffic accidents. Some potential limitation of this system includes the high power consumption of smartphone traffic accidents system, which may result in reduced battery life. The increased possibility of false alarms due to filters used in smartphones was also identified as a limitation for minor accidents such as sideswipe collisions when a car is in low-speed traffic [14]. Other studies have proposed the use of more specific sensors, such as airbag systems, for accident detection. However, these systems are limited to detecting accidents in specific types of cars. As such, there is a need for a more generic system that can be utilized across all types of vehicles [9]. To address the limitations of previous studies, Mateen et al. [12] proposed a framework for an Accident Alert and Light and Sound System (AALS) using pre-saved locations to reduce the time needed to locate accidents without relying on GPS. The framework attempts to reduce the risk of Multiple vehicle collisions by providing real-time notifications to nearby vehicles and emergency services. The use of various sensors for accident detection is a promising area of research that can improve the safety of roads and reduce the occurrence of MVCs.

C. Social Network and Geosocial Media Data

The enormous amount of information being constantly shared daily across various social media platforms contains artifacts that can be analyzed to generate meaningful insights for traffic events [15]. Although the ability to monitor and analyze the exploding information manually seems impossible based on the high volume and unstructured formats of the information being presented [16]. Monitoring traffic-related information on social media has been proven beneficial in detecting traffic events. Xu et al. [17] provided a synthesis of research work that explored the usage of geosocial media data for detecting traffic events. Events such as road accidents, road closures, and traffic conditions are typically shared among people through social media platforms. Such events can be tracked down with the aid of GPS in getting first responders to the event location and often contain information that triggered such events. Xu et al. [18] proposed framework achieved 81% accuracy in classifying Twitter data into non-traffic events, traffic accidents, roadwork, and severe weather conditions. Similarly, Salas et al. [19] developed a framework leveraging social media data to crawl, process, and filter social media data for implying traffic incidents and real-time detection of traffic events with text classification algorithm [20].

III. PROLIFERATION OF TRAFFIC ACCIDENT EVENT IN U.S

This section analyzes traffic accidents using data from the National Highway Traffic Safety Administration (NHTSA) Crash Report Sampling System (CRSS). This aims to understand better the factors contributing to traffic accidents

and identify potential strategies to improve highway safety. Motor vehicle crashes continue to be a significant public health concern in the United States, resulting in over 38,000 deaths and millions of injuries each year [21]. To reduce these numbers, the NHTSA has been collecting crash data since the 1970s using various data collection systems, including the Crash Report Sampling System (CRSS), Fatality Analysis Reporting System (FARS), and Crash Investigation Sampling System (CISS). The Crash Report Sampling System (CRSS) is a compilation of crashes documented by law enforcement officers, encompassing all modes of transportation, including cars, bikes, and pedestrians. This database offers a broad perspective of accidents, covering everything from minor accidents to fatal crashes. The information obtained through the CRSS is analyzed to pinpoint areas of concern regarding road safety, track changes over time, and evaluate the effectiveness of road safety initiatives and regulations [21]. In this section, we aim to analyze the CRSS data to identify factors contributing to traffic accidents and develop potential strategies to improve traffic safety. The dataset includes information on various factors related to each crash, including the location, date and time, type of vehicles involved, driver characteristics, and weather conditions, among others, within five years (2016 -2020). The CRSS sample is selected from 60 representative regions across the United States, which are chosen based on factors such as geographical location, population density, miles driven, and the frequency of traffic crashes. The sample is drawn randomly from the approximately 6 million traffic accidents reported annually. Figure 1 presents the trend analysis of

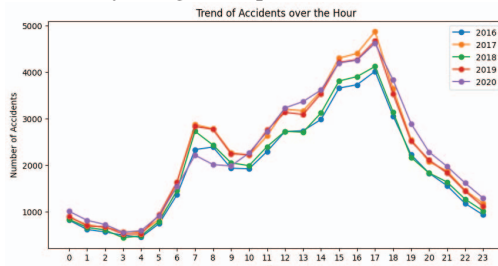


Fig. 1. Hourly Trend of Accident (2016-2020).

traffic accidents from 2016 to 2020 by hour. The data shows a consistent increase in accidents every hour, with a noticeable spike between 7 am and 8 pm. However, this spike was reduced in 2020. Additionally, the year 2017 significantly increased accidents between 1 pm and 6.30 pm. In 2020, traffic accidents were higher than in any other year, particularly from 6 pm to 3 am.

These results suggest that certain times of day may be more susceptible to accidents, which could be due to factors such as increased traffic volume, driver fatigue, or other behavioral issues. Reducing the spike in early hours in 2020 may indicate the positive impact of preventative measures and educational campaigns. However, the overall trend of traffic accidents has been increasing, as demonstrated by the higher number of accidents in 2020 compared to previous years. This highlights the importance of continued attention and efforts toward improving road safety and reducing the number

of accidents. The study by Fors et al. [22] analyzed nighttime traffic in Sweden and found that 25% of personal injury traffic accidents and 29% of all fatalities occurred during the late evening (dark hours). To address these issues, physical structures such as street lighting, well-designed roads, traffic signals, and technical solutions such as accident detection systems and weather warning signals can be implemented to reduce the number of traffic accidents during night-time. Furthermore, Figure 1 demonstrates it is essential to consider that human vision is less effective in darkness, highlighting the need for additional measures to ensure road user safety during these times.

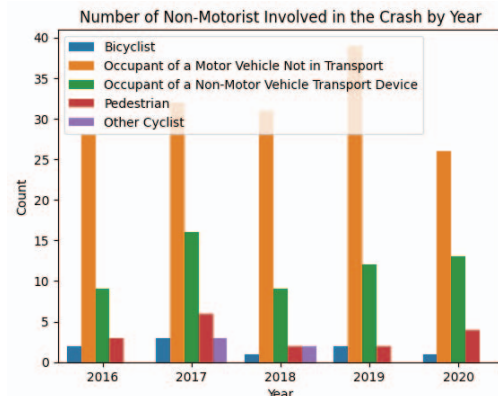


Fig. 2. No of Non-Motorist involved in Car Crash (2016-2020).

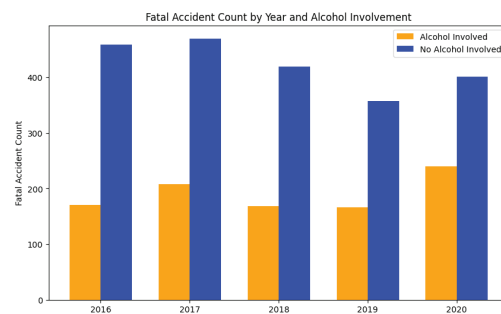


Fig. 3. Fatal Accidents with Alcohol Involvement.

Figure 2 presents the number of nonmotorists involved in traffic accidents. The CRSS manual defines nonmotorists as individuals who are not operating a motor vehicle, including pedestrians, cyclists, occupants of motor vehicles that are not in transport, people associated with nonmotorist conveyances such as baby carriages or skateboards, or individuals outside of a traffic way or in a house who are involved in a traffic accident. Most traffic accidents involving nonmotorists involve people in other vehicles not in transport affected by the crash, followed by occupants of non-motor vehicle transport devices. There was a spike in the number of pedestrians involved in traffic accidents in 2017, followed by a significant increase in 2020. On the other hand, the number of cyclists involved in traffic accidents significantly decreased between 2017 and 2019. These findings align with previous research on the vulnerability of nonmotorists in traffic accidents. The study of Justin Tyndall [23] found that pedestrian fatalities accounted for a disproportionate number of traffic deaths in

the united states. One hundred thousand pedestrians death were recorded out of the 741,000 fatalities recorded in the US between 2000 and 2019. Measures such as pedestrian-friendly infrastructure and education campaigns could help to reduce pedestrian accidents [24]. Bicyclists involved in crashes with motor vehicles are more likely to sustain severe injuries. Another study by National Transportation Safety Board [25] found that using bicycle-friendly infrastructure, such as dedicated bike lanes and education campaigns aimed at cyclists and drivers, could help reduce the number of bicycle accidents. Considering nonmotorists' safety in traffic is crucial as they are often more vulnerable to severe injury or death in the event of an accident. Figure 3 shows traffic fatalities related to alcohol consumption. The NHTSA indicates that the number of traffic accidents involving fatalities related to alcohol consumption was higher in 2020 compared to other years accounting for a 14% increase from 2019, with 11,654 fatalities recorded in 2020 due to alcohol-impaired driving [1]. The research of Smailovića et al. [26] also correlated driving under the influence of alcohol with excessive speed, driving at night-time, and failing to wear seat belts [26]. Different

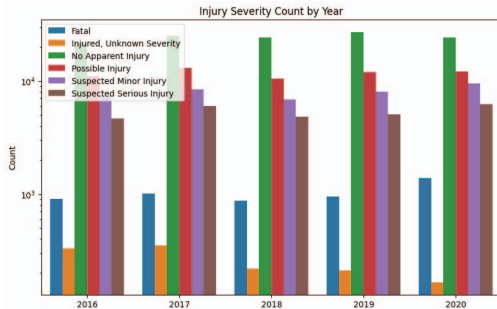


Fig. 4. Severity of Injury.

types of injury severity in traffic accidents across the five-year period from 2016 to 2020 are shown in Figure 4. There is a noticeable increase in the number of fatal injuries over time, with the highest number of fatal injuries recorded in 2020. A significant number of suspected injuries and minor injuries were also observed in all five years. Lin et al. [27] investigated the environmental factors associated with severe crash injury in Taiwan by linking clinical data with police reports. The study found that drunk driving, early morning driving, flashing signals at intersections, and single-motorcycle crashes were identified as risk factors for severe injury. Interestingly, the research indicated that adverse weather conditions do not significantly increase the risk of traffic accidents which is in contrast with the analysis of fatal accidents in different regions in the United States and associated weather conditions shown in Figure 5. The authors suggested that a better understanding of traffic patterns during early morning hours and the use of flashing signals at critical intersections could help to reduce the number of traffic accidents. Additionally, they found that the time elapsed between the incident and medical care (i.e., the time to the hospital) is a crucial factor in reducing injury severity. This suggests that the implementation of spontaneous emergency response systems for accident detection could help

to save more lives and reduce reporting errors [6], [27]. The

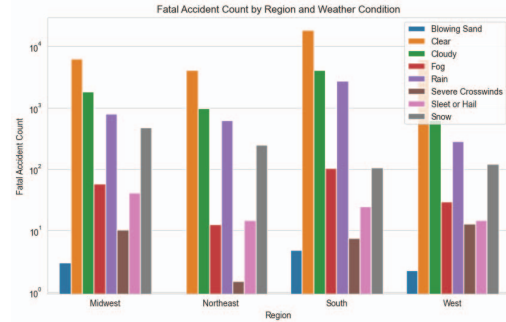


Fig. 5. Fatal accident in different regions across the US and weather conditions associated.

research of Liu et al. [28] investigated the factors contributing to traffic accidents in low-light conditions in China using a predictive model. Key findings from the research indicate that several factors, such as road conditions, weather conditions, median dividers, and road surface, contribute to fatal traffic accidents. The findings also showed that the probability of fatal accidents is 2.38 times higher at intersections [28]. The analysis in Figure 5 also illustrates that weather conditions such as cloudy, rain, and snow contribute to fatal accidents in the United States, with the midwest region primarily impacted.

IV. METHODOLOGY

The transportation industry has experienced significant growth in implementing computer vision systems. These systems have proved to be instrumental in enhancing traffic safety, mitigating traffic congestion, and improving transportation efficiency [29]. Among the various computer vision systems, action recognition has been identified as a crucial area of focus in the transportation industry, specifically in autonomous transportation and road traffic safety management. Adopting autonomous transportation systems in smart cities can immensely benefit from cutting-edge systems such as radar detection, wrong-way detection, license plate recognition, and road construction navigation, all of which can be enhanced by developing advanced algorithms and technologies. The continued advancement of action recognition technologies in the transportation industry is essential to improve efficiency and safety. Huang et al. [30] propose an intelligent intersection system that utilizes two-stream convolutional networks to detect near-accidents in real-time traffic video. The system combines spatial and temporal information to improve the accuracy of detecting near accidents. The experimental result of the proposed system reported high accuracy in the real-time detection of near-accidents with the potential to enhance road safety by alerting drivers and reducing the occurrence of accidents at intersections. Zhao et al. [31] proposed a novel approach in temporal action detection by predicting actions' start and end times in untrimmed videos. The SSN framework demonstrates competitive performance in both action detection and classification tasks, showcasing the potential to recognize and accurately classify actions in untrimmed videos. Similarly, Lee et al. [32] proposed weakly-supervised temporal action localization to detect intervals of action instances with only

video-level action labels and separate background frames that do not contain an action class. The proposed model was tested on the Thumos dataset using the I3D architecture, the feature extraction process involved splitting videos into multi-frame non-overlapping segments to mitigate videos with varied lengths. The extracted features were then embedded and fed into a 1D convolution layer with ReLU activation. Segment-level class scores were predicted for the embedded features, which were used for action localization. The model was optimized using three losses: action classification losses, uncertainty modeling loss, and background entropy loss. This method can be used to improve action recognition models in real-world scenarios and improve such models’ accuracy.

V. PROPOSED FRAMEWORK

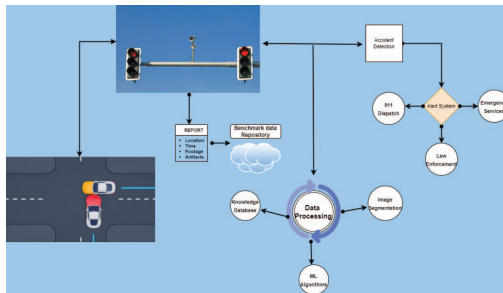


Fig. 6. Proposed Framework for Autonomous Accident Detection System.

The proposed accident detection framework for autonomous transportation focuses on three key elements: Traffic Camera Utilization, Data Processing, and Accident Detection. These components work synergistically to create a comprehensive and efficient system for identifying and responding to traffic accidents.

- **Traffic Camera Utilization:** Existing traffic surveillance cameras can be repurposed to perform sophisticated tasks such as accident detection and monitoring traffic violations. These cameras continuously capture valuable information such as location, time of the accident, accident footage, and other artifacts.
- **Data Processing:** The data extracted from the traffic cameras undergo a series of processing steps, including image segmentation and application of machine learning algorithms. These algorithms work with the knowledge database to analyze the data and identify accident-related patterns. The output generated from data processing is then sent to the Accident Detection System for further evaluation and decision-making.
- **Accident Detection and Alert System:** The Accident Detection System comprises three integral components that revolve around a decision alert system. These components work in tandem to detect accidents promptly. Upon identifying an accident, the alert system initiates the following actions: 1) 911 Dispatch Call: The system automatically places a call to the 911 emergency dispatch center, providing essential information such as location, time, and nature of the accident. This ensures that first responders are notified promptly and can reach the scene

as quickly as possible. 2) Emergency Services Notification: In parallel to contacting the 911 dispatch center, the alert system also notifies other relevant emergency services. 3) Law Enforcement Communication: Lastly, the system alerts law enforcement agencies about the accident, enabling them to manage traffic flow, investigate the incident, and take appropriate legal action as needed.

This paper presents a novel framework for accident detection in autonomous transportation systems that integrates traffic cameras, data processing, and machine learning algorithms. By automating the alert system and coordinating necessary emergency services, this framework can improve road traffic safety and reduce the severity of traffic accidents. Previous studies have approached traffic accident detection from different angles, including clustering, fuzzy logic, K-Means, and deep learning models. However, these methods often create computational and cost constraints because the developed model may need to be more generalizable to other areas or traffic routes not included in the data collection. This research proposes a robust benchmark dataset generalizable to various weather conditions and road structures to address this challenge. The proposed framework builds upon the DSTGCN framework proposed by Le et al. [33], which was efficient in predicting accidents at the road level by modeling the spatial correlations and temporal dependencies of the road structure. Our research presents a promising solution for accident detection in autonomous transportation systems and contributes to developing a safer and more efficient transportation system.

VI. CONCLUSION AND DISCUSSION

Our analysis of traffic accidents from 2016 to 2020 highlights the continued proliferation of traffic accidents and the urgent need for practical solutions to reduce the number of accidents and improve road safety. The findings of this study indicate that the number of traffic accidents has been increasing over the years, with a noticeable spike in the number of accidents in the early morning and late evening. One of the significant trends in our analysis is the increase in the number of fatal accidents. The number of fatal accidents has steadily increased over the years, with the highest fatalities recorded in 2020. This also underpins the importance of developing effective strategies for reducing the number of fatal accidents and improving road safety. Our analysis of traffic accidents is consistent with the research findings of other studies on the impact of various factors on traffic safety, including light conditions, weather conditions, road design, and driver behavior. Advanced technologies, such as autonomous accident detection systems, can play a crucial role in reducing the number of accidents and improving road safety by providing real-time data on the severity of accidents and improving the speed and efficiency of emergency response, which has proven to reduce accident severity. Furthermore, the analysis of fatalities related to alcohol consumption and the severity of injuries highlights the need for continued attention to reducing driving under the influence of alcohol and improving emergency response. The findings from the

research on the performance appraisal of urban lighting systems suggest that the positioning of electronic billboards, the presence of oncoming vehicle lights, and poor lighting during inclement weather can affect the performance of urban street lighting systems and contribute to the number of accidents. Our proposed framework utilizes Action Recognition (AR) for accident detection, which can provide a valuable tool for detecting accidents when they occur and providing real-time updates to other motorists. Technological progress and logical reasoning in developing innovative solutions utilizing existing technologies are imperative to enhance public safety and minimize traffic accidents. With the aid of these technologies, law enforcement officials will be able to investigate accident scenes and conduct thorough investigations. In conclusion, this study highlights the importance of continued research and attention to improving road safety and reducing the number of traffic accidents. Advanced technologies, such as autonomous accident detection systems, can be crucial in this effort. However, further research is needed to fully understand the reasons behind the trends in traffic accidents and determine the most effective strategies for reducing the number of accidents. The elapsed time between an accident and medical emergency attendants is also a crucial factor in reducing the severity of accidents, and the proposed framework provides an opportunity to understand the best strategies for improving emergency response times.

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